

Deep Learning-Driven Insights into Enzyme–Substrate Interaction Discovery

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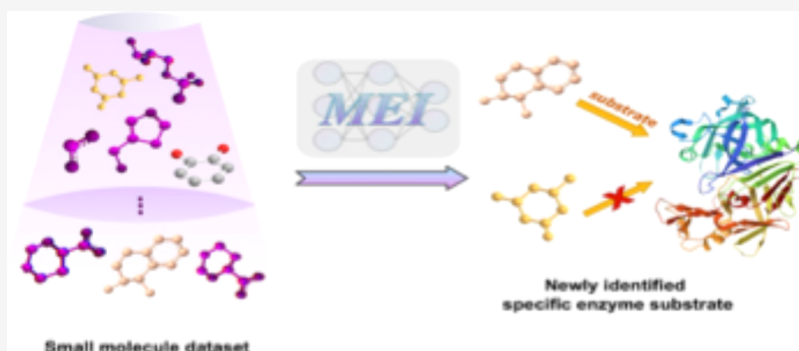
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ABSTRACT: Enzymes are ubiquitous catalysts with enormous application potential in biomedicine, green chemistry, and biotechnology. However, accurately predicting whether a molecule serves as a substrate for a specific enzyme, especially for novel entities, remains a significant challenge. Compared with traditional experimental methods, computational approaches are much more resource-efficient and time-saving, but they often compromise on accuracy. To address this, we introduce the molecule–enzyme interaction (MEI) model, a novel machine learning framework designed to predict the probability that a given molecule is a substrate for a specified enzyme with high accuracy. Utilizing a comprehensive data set that encapsulates extensive information on enzymatic reactions and enzyme sequences, the MEI model seamlessly combines atomic environmental data with amino acid sequence features through an advanced attention mechanism within a hierarchical neural network. Empirical evaluations have confirmed that the MEI model outperforms the current state-of-the-art model by at least 6.7% in prediction accuracy and 8.5% in AUROC, underscoring its enhanced predictive capabilities. Additionally, the MEI model demonstrates remarkable generalization across data sets of varying qualities and sizes. This adaptability is further evidenced by its successful application in diverse areas, such as predicting interactions within the CYP450 enzyme family and achieving an outstanding accuracy of 90.5% in predicting the enzymatic breakdown of complex plastics within environmental applications. These examples illustrate the model’s ability to effectively transfer knowledge from coarsely annotated enzyme databases to smaller, high-precision data sets, robustly modeling both sparse and high-quality databases. We believe that this versatility firmly establishes the MEI model as a foundational tool in enzyme research with immense potential to extend beyond its original scope.

INTRODUCTION

In the realm of modern drug development and biomanufacturing, a profound understanding of enzyme-catalyzed reaction mechanisms is pivotal.¹ Enzymes catalyze specific chemical reactions with remarkable efficiency, substantially enhancing reaction rates and thereby playing an essential role in numerous biochemical processes.^{2–5} However, characterizing enzyme–substrate interactions remains a significant challenge, as it largely depends on experimental methods that are expensive and time-consuming.^{6,7} Despite their effectiveness, these experimental methods still face substantial challenges when confronted with a large number of unidentified enzymes and substrates.^{8,9} Although the UniProt database contains over 36 million enzyme entries, over 99% lack high-quality catalytic

reaction annotations.¹⁰ While high-throughput experimental methods for determining enzyme–substrate relationships have continuously made progress, these approaches are still in their nascent stages. Furthermore, even high-throughput methods confront limitations due to the enormous search space of potential small-molecule substrates, limiting experimenters to physically probing only a small subset. Computational

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approaches, such as QM/MM simulations, present viable alternatives,¹¹ but they incur substantial computational costs and are constrained by algorithmic limitations. While progress in sequence-based functional prediction and protein design algorithms^{12,13} has been steady, there has been no significant breakthrough yet, as this approach heavily relies on the empirical knowledge of scientists.

The advent of the big data era and the rise of computational biology have opened new vistas for enzyme research.^{14,15} Gradually, data-driven models, particularly metabolic network models informed by metabolomics and genomics data, are being integrated into the traditional design-build-test-learn (DBTL) cycle, thus accelerating research and reducing costs.¹⁶ However, current methodologies still struggle to efficiently manage the vast amount of unknown enzymes and substrates, and accurately predicting enzymes that catalyze specific reactions remains a formidable challenge.^{17,18}

Despite the availability of various methods and platforms for exploring enzyme structures and functions, accurately identifying specific substrates for enzyme catalysis remains a formidable task. The current scarcity of enzyme-specific reaction data poses significant obstacles in relying solely on known reactions for enzyme identification. While machine learning-based tools have exhibited promise in this field, most are specifically designed for a limited range of enzyme families with extensive training data or correlate substrates with EC categories, rather than targeting specific enzymes. For instance, Faulon et al.¹⁹ introduced a method that concurrently considers reactions and enzymes to assess catalytic potential. They characterized both reactions and enzymes using the atomic environment of the reaction center and amino acid sequences and trained a Gaussian process classification model to identify enzymes in *Escherichia coli* associated with *N*-acetyl-L-leucine synthesis. Mou et al.,¹⁸ on the other hand, developed a machine learning model for predicting substrates of bacterial nitrile hydratases using the features from the enzymes' three-dimensional structure and active site. A similar approach was adopted by Yang et al.²⁰ for predicting the substrate range of plant glycosyltransferases. These models are typically focused on specific enzymes or families and rely heavily on the comprehensiveness of the experimental data, which is expected to be a challenge for most enzyme families. Goldman et al.'s study²¹ on the applications of machine learning in predicting enzyme–substrate interactions across various enzyme families emphasized the limitations of directly adopting standard compound–protein interaction (CPI) modeling strategies. They observed that compared to purpose-built enzyme–substrate interaction models, CPI models often yield inferior results. Their findings suggest that although enzyme–substrate interaction predictions are similar to CPI predictions,^{22–24} the proteins used in CPI model training often lack enzymatic functions and the molecules used are typically not substrates. Our subsequent findings further clarified that models initially trained on BindingDB or Davis data sets,^{25,26} still performed suboptimally when retrained on the ESP data set for enzyme–substrate interaction prediction. This suggests that predicting enzyme–substrate interactions presents a significant challenge for traditional CPI models as it falls outside their typical scope. This is potentially due to the unique structural dynamics of enzyme–substrate interactions, which involve both sequence specificity and the three-dimensional active site, making CPI models based solely on sequence data less effective. Additionally, CPI models struggle to account for enzyme promiscuity

and multifunctionality. As confirmed by our own experiments, these factors contribute to the suboptimal performance of CPI models in this context.

Recently, Kroll et al. introduced the ESP model,²⁷ a multifaceted machine learning approach that significantly advanced this field, achieving over 91% accuracy in predicting enzyme–substrate pairs across diverse data sets. While this model demonstrated its effectiveness for a broad range of enzymes and metabolites in its training data, surpassing models designed for specific, well-studied enzyme families, our further evaluations revealed its limited generalization capabilities, particularly in accurately predicting enzymes or molecules unseen in its training data set. This limitation underscores a significant challenge for the application of the ESP model in more complex real-world scenarios, where new enzymes and substrates are frequently encountered. To address this challenge, it is crucial to refine and expand the model's training data set, and then to develop new models designed inherently for greater adaptability and generalization. These new models should integrate advanced machine learning techniques capable of learning a more diverse range of enzyme–substrate interactions, including those that are less common or entirely novel. Thus, the pursuit of new modeling approaches, coupled with the expansion of existing data sets, will be pivotal in overcoming current limitations and unlocking the full potential of machine learning in enzyme–substrate interaction prediction. This progress is crucial for translating these computational predictions into practical, real-world applications in the biomedical and biotechnological fields.

In this study, we curated a comprehensive data set that contains a vast array of enzyme-catalyzed reactions and the corresponding enzymes, providing unprecedented insights into enzyme–substrate interactions. Based on this robust foundation, we introduced the MEI model, an innovative machine learning framework specifically designed to predict the probability that a given molecule is a substrate for a specified enzyme. This model focuses on determining the likelihood of interaction, thereby streamlining the identification process for potential substrates. The MEI model leverages a comprehensive data set that includes extensive enzyme-catalyzed reactions and enzyme sequences. It integrates atomic environmental data with amino acid sequence features by using an advanced attention mechanism within a hierarchical neural network architecture, which is specially designed for accurate enzymatic function prediction. A key feature of the MEI model is its foundational design, allowing for fine-tuning to predict additional enzymatic properties beyond the substrate probability. This adaptability enables the model to be customized for specific applications, such as predicting interaction specifics within enzyme families such as CYP450 and their corresponding small molecules.

Beyond the realm of biomedicine, the MEI model has demonstrated its efficacy in green chemistry, particularly in predicting enzyme activity for the degradation of diverse plastics, rivaling state-of-the-art (SOTA) benchmarks. Its exceptional predictive capability in identifying unexplored enzyme–substrate interactions holds significant potential to revolutionize synthetic biology by optimizing enzymatic pathways, thus enabling more sustainable and eco-friendly chemical processes.

Our exploration of the MEI model in the realm of enzyme-catalyzed reactions provides biochemists with a powerful computational tool for identifying novel enzyme–substrate

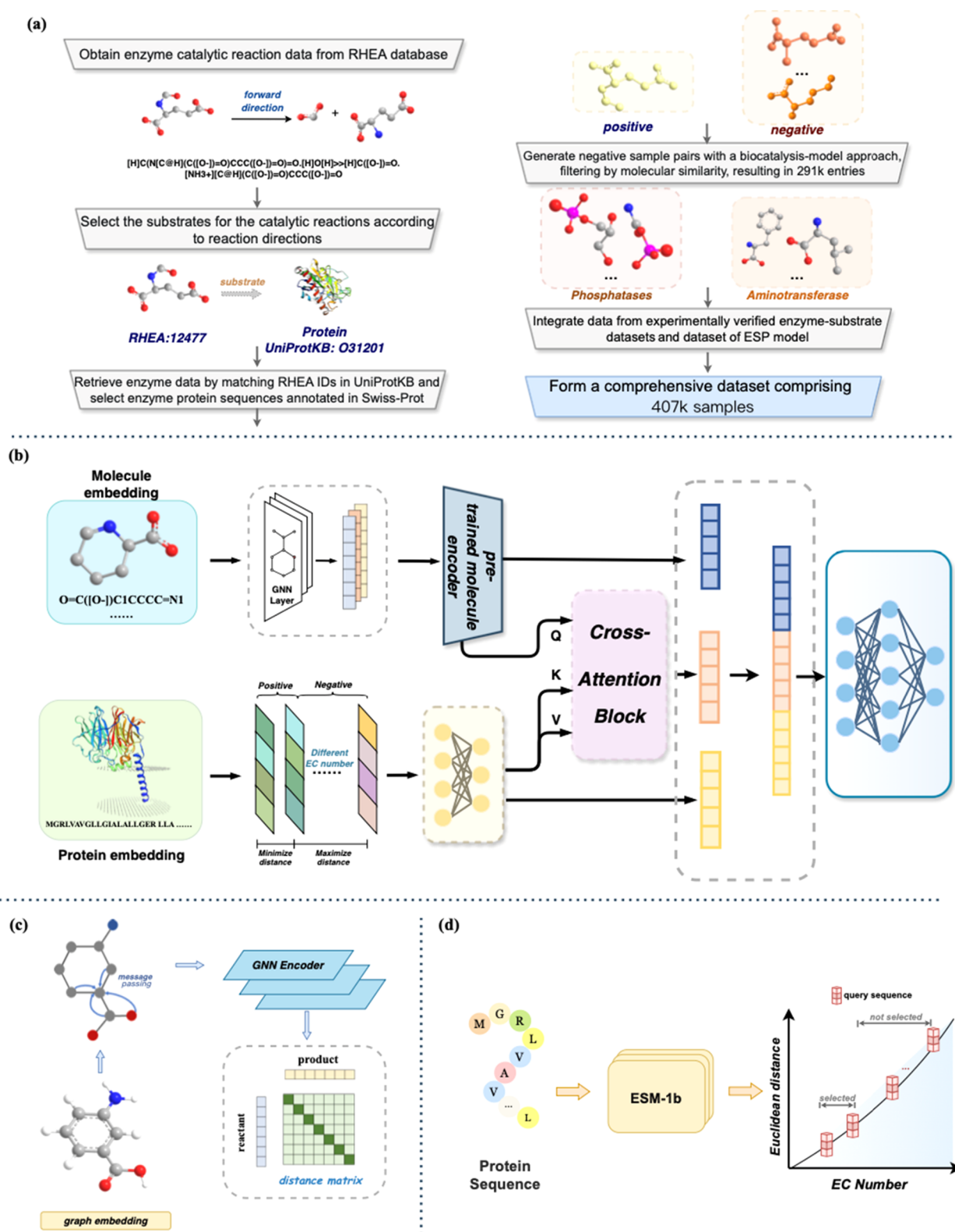


Figure 1. Overview of the MEI model and the workflow involved in the study of enzyme–substrate interactions. (a) Workflow for data set compilation and preprocessing in enzyme–substrate interaction study. The process includes the collection, cleaning, and preprocessing of relevant data sets to ensure high-quality input for subsequent analysis. (b) Network architecture of the MEI system. The system architecture includes modules for learning deep representations of substrates and enzymes, capturing features of their interactions, and integrating these features to make predictions. (c) A GNN encoder processes molecules and uses substrate–product pairs for the specification of molecular representation constraints. The graph neural network (GNN) encoder captures the molecular structures and ensures an accurate representation by leveraging substrate–product pairs. (d) Contrastive learning-based framework of the protein encoder. When used as a classification model, maximum separation is implemented to prioritize confident predictions of Enzyme Commission (EC) numbers based on the ranking order.

interactions, which can guide the discovery and engineering of new enzymes. By accurately predicting potential substrates for existing enzymes, the MEI model highlights unexplored biochemical pathways and reaction mechanisms that experimental researchers can investigate further. This highlights the model's substantial potential and versatility, serving as a catalyst for advancements in biochemistry and beyond, driven by the infusion of machine learning innovations.

METHOD

Data Sets. We queried enzyme-catalyzed reaction data from UniProt protein sequence database,¹⁰ which utilizes Rhea's chemical reaction descriptions to manage reactions at the specific enzyme level and standardize reactant names. Consequently, we downloaded relevant reactions from the Rhea database,²⁸ managed by the Swiss Bioinformatics Institute, due to its extensive, high-quality enzyme annotations seamlessly integrated with UniProt. This integration ensures accuracy and consistency in analyzing enzyme-molecule relationships. Rhea's reactions, curated from peer-reviewed literature, include enzyme-catalyzed and spontaneous reactions, described using ChEBI chemical dictionaries for computational tractability. Rhea also covers IUBMB-classified and supplementary reactions, making it ideal for enzyme annotation, metabolic modeling, and omics analyses.

Our initial data processing, illustrated in Figure 1a, began with a focus on entries with UniProt IDs. Subsequently, we transformed the gene names of our target enzymes into UniProtKB/Swiss-Prot IDs by utilizing the Retrieve/ID mapping module of UniProt, ensuring high reliability and minimal redundancy. For enzymes with matching UniProt IDs, we retrieved their relevant protein sequence FASTA files from UniProt. We then categorized the reactants/products in this data set as substrates based on the enzyme's catalytic reaction direction, treating them as positive samples. Each enzyme's multiple substrates/products were then split into one-to-one enzyme–substrate pairs. The resulting data set contains 66,724 such pairs, marking a significant step forward in annotating enzyme–substrate interactions within enzyme catalysis. By establishing a standardized framework for such annotations, this data set serves as a foundational resource, facilitating advancements in research and applications in this critical field.

To simulate real-world prediction scenarios and address the scarcity of comprehensive data for nonenzyme–substrate pairs, we utilized the biocatalysis model,²⁹ a predictive tool based on the Molecular Transformer architecture,³⁰ to generate negative samples for training. Specifically, for each known enzyme–substrate pair, the biocatalysis model was used to generate 10 predicted reactant and product molecules, resulting in a data set where the number of nonsubstrate small molecules is 10 times greater than that of actual enzyme reaction substrates. These enzyme–nonsubstrate pairs were then added as negative examples to complement the existing positive samples, thereby enhancing the training process and predictive performance of the MEI model. The biocatalysis model achieved a top-1 accuracy of 49.6% in forward reaction predictions and a 39.6% top-1 single-step round-trip accuracy for retrosynthetic pathways.

To reduce the bias stemming from the limited data set, we calculated the RDkit fingerprint similarity between the predictions and actual substrates. Consequently, we selected molecules with RDkit fingerprint similarities falling within the

range of 0.6–0.95 to serve as negative control substrates for the corresponding enzymes in our model.

Furthermore, we compiled high-throughput enzyme activity screening data from various studies, encompassing amino acid sequences of diverse small enzyme families along with their substrates and nonsubstrates' SMILES sequences.^{31,32} After integrating this data into the ESP model's training data set and eliminating duplicates, our refined data set comprised over 407,000 samples. We employed an enzyme similarity-based split method using the Levenshtein distance for protein sequence comparison to ensure at least 60% sequence divergence between enzymes in the training and test sets. The final data set was divided into a training set (90%) and a test set (10%).

Data Representation. In our model, the primary inputs consist of protein sequences and Simplified Molecular-Input Line-Entry System (SMILES) representations of compounds. For compounds, we utilized a graph neural network equipped with TAGConv layers.³³ TAGConv is a message-passing graph convolutional network that updates the features of each node by analyzing the surrounding neighborhood information. This method, coupled with reaction constraints, efficiently transforms SMILES representations into detailed molecular diagram representations. For protein sequences, we initially converted amino acid strings into numerical vectors employing the pretrained module CLEAN,³⁴ effectively vectorizing the protein sequences for subsequent processing in our model.

Model Architecture. Figure 1b shows the framework of MEI, consisting of three core modules: an embedding module, a representation learning module, and an interaction prediction module. The representation learning module encompasses two submodules: a molecular graph aggregator and a protein encoder. The protein encoder uses the CLEAN algorithm, rooted in Evolutionary Scale Modeling (ESM), to represent proteins. It integrates enzyme representation and utilizes multilayer deep neural networks (DNNs) to independently learn deep representations of enzymes and substrates. The interaction module is dedicated to capturing the substructural interactions between enzyme–substrate pairs. Finally, deep representations of enzyme–substrate pairs, along with their interaction features, are fed into an output module to produce predictive results.

Protein Feature Encoder. The rapid evolution of DNA sequencing technology has significantly expanded the availability of protein sequences, with the UniProt database now housing an extensive collection of approximately 190 million protein sequences.¹⁰ However, a mere 0.3% (about 500,000) of these sequences have been subjected to manual review, and only 19.4% are supported by substantial experimental evidence.³⁴ This scarcity of verified data has necessitated a heavy reliance on computational methods for protein function annotation. Yet, large-scale, community-driven studies, such as the Critical Assessment of Functional Annotation (CAFA), indicate a significant margin of error in these computational tools, with approximately 40% of automated enzyme annotations found to be inaccurate.³⁵ Therefore, accurate functional annotation of proteins remains a significant and complex challenge in protein science.³⁶

Recently, Yu et al. developed a machine learning algorithm named CLEAN (Contrastive Learning enabled Enzyme Annotation),³⁴ which revolutionizes the prediction of enzyme functions in unexplored enzymes with unprecedented precision and sensitivity. Trained on high-quality data from UniProt,

CLEAN takes amino acid sequences as the input and generates a probability-ranked list of enzyme functions (EC numbers) as the output. By leveraging extensive evolutionary data, CLEAN captures both local and long-range interactions within proteins, therefore enhancing the prediction accuracy in computational biology. This nuanced understanding enables CLEAN to discern diverse enzymes with greater efficacy and deliver precise annotations of their EC numbers.

Our approach builds on the foundation established by CLEAN, utilizing the ESM-1b language model³⁷ for protein sequence encoding, as illustrated in Figure 1d. ESM-1b, a state-of-the-art pretrained model, employs transformer architectures to generate contextual embeddings of protein sequences, capturing both local and long-range interactions with high fidelity. Inspired by CLEAN, we adopted a contrastive learning framework to delineate the representation space of enzymes using the Euclidean distance metric, enabling effective discrimination among various enzymes. Leveraging these insights, we further refined our protein encoders to improve enzyme characterization tasks.

Specifically, during pretraining, we integrated EC classification information to enhance the model's ability to differentiate between various enzyme types. A Gaussian Mixture Model (GMM) was incorporated into our framework for uncertainty estimation. The GMM captures the distribution of embedding distances between protein sequences and their corresponding EC numbers, thereby providing confidence scores for the predictions. The two components of the GMM characterize the intra-EC and inter-EC number embedding distance distributions. During enzyme function prediction, the GMM's probabilistic output guides the prediction process, ensuring that specific EC numbers are only predicted when the associated confidence score is sufficiently high, thereby mitigating overprediction and improving overall prediction precision. In the fine-tuning stage, we modified the network's output layer, resulting in a refined, function-aware 128-dimensional sequence embedding of the input enzyme. This embedding effectively streamlines downstream enzyme characterization tasks.

Moreover, the contrastive loss functions included Triplet Margin Loss and Supcon-Hard Loss. The Triplet Margin Loss is defined as

$$\mathcal{L}_{\text{TM}} = \|z_a - z_p\|^2 - \|z_a - z_n\|^2 + \alpha$$

where z_a represents the embedding of the anchor enzyme, z_p is the positive embedding from the same EC class, and z_n is the negative embedding from a different EC class, with α set to a margin of 1.

The Supcon-Hard Loss extends this approach by sampling multiple positives and negatives, and is defined as

$$\mathcal{L}_{\text{sup}} = \sum_{e \in E} \frac{-1}{|P(e)|} \sum_{z_p \in P(e)} -\log \frac{\exp(z_e \cdot z_p / \tau)}{\sum_{z_a \in A(e)} \exp(z_i \cdot z_a / \tau)}$$

where τ is the temperature parameter, set to 0.1. The summation iterates over all enzymes e in set E , with $P(e)$ denoting the set of positive samples and $A(e)$ denoting the set of available negative samples for enzyme e . This loss function normalizes the enzyme embeddings, balancing the influence of positive and negative samples during training, which significantly enhances performance, especially on data-scarce data sets.

By integration of these advanced techniques, our protein feature encoder produces semantically rich protein embeddings that, coupled with contrastive learning and uncertainty estimation, markedly improve the accuracy and robustness of enzyme function annotation.

Compound Feature Encoder. Molecular Representation Learning (MRL) methodologies are designed to embed molecules into multidimensional real vector spaces. In our study, we employed Graph Neural Network (GNN)-based MRL technique to encode substrate molecules, augmented by the inclusion of chemical reactions to enhance the learning of molecular representations.^{38–40} The fundamental concept is to maintain the equivalence or correspondence between substrates and products within the embedding space. This involves ensuring that for each enzyme-catalyzed reaction, the combined embedding of the reactants aligns with that of the products. This constraint significantly enhances the generalization capability of the molecular embeddings, optimizing their performance in diverse downstream applications.⁴¹

A key innovation in our molecular embedding approach is the enforcement of a balance between the sum of the reactant embeddings and that of the product embeddings for each chemical reaction, formalized as

$$\sum_{s \in S} h_s = \sum_{p \in P} h_p$$

where S and P represent the sets of substrates and products, respectively, and h_s and h_p denote their corresponding embeddings. This constraint effectively organizes the embedding space, ensuring that molecules involved in chemical reactions are appropriately represented relative to one another, thereby enhancing the model's generalization ability.

To further enhance the embeddings, we adopted a contrastive learning framework, drawing inspiration from the work of Wang et al.⁴² This framework encourages similar embeddings for correctly matched reactant–product pairs (positive pairs) and dissimilar embeddings for incorrectly matched pairs (negative pairs). The contrastive loss function is defined as

$$\mathcal{L}_B = \frac{1}{|B|} \sum_i \left\| \sum_{s \in S_i} h_s - \sum_{p \in P_i} h_p \right\|^2 + \frac{1}{|B|(|B| - 1)} \sum_{i \neq j} \max \left(\gamma - \left\| \sum_{s \in S_i} h_s - \sum_{p \in P_j} h_p \right\|^2, 0 \right)$$

where γ is a margin hyperparameter and B is a minibatch of reactions. This approach effectively differentiates consistent substrate-product pairings from inconsistent ones, enhancing the model's learning efficiency.

This contrastive learning paradigm⁴³ efficiently aids in obtaining meaningful embeddings for enzymatic reactions. After the GNN is pretrained with reaction information, the model no longer requires access to reaction data to generate substrate representations. The trained GNN can produce high-quality embeddings for substrates, capturing both local and global molecular features. As illustrated in Figure 1c, contrary to traditional models that merely describe molecular features, this innovative model integrates key information from enzymatic reactions, thereby bolstering its analytical capa-

bilities and providing an in-depth understanding of the dynamic interplay between enzymes and substrates.

Substrate–Enzyme Interaction Module. This model aims to enhance predictions of substrate–enzyme interactions by capturing nuanced interaction patterns. It integrates a cross-attention block,^{44,45} which fuses profound characterizations of molecules and enzymes from molecular and amino acid encoders, respectively. Such integration is essential for understanding the specific interactions within each substrate–enzyme pair, moving beyond basic classification tasks that rely solely on the individual substrate and enzyme properties. Initially designed to enhance machine translation quality by aligning diverse representations (e.g., source and target languages), the attention mechanism in this context enables neural models to effectively consider remote feature relationships, thus improving the task performance and offering interpretability in model predictions. Ablation studies (Table 2) demonstrate that the attention module significantly improves performance, highlighting its crucial role in effectively modeling enzyme–substrate interactions. The efficacy of the attention mechanism in generating sophisticated representations has been validated in numerous studies focused on drug–target interaction predictions.^{24,46}

In our research, the cross-attention mechanism harmoniously combines the sequential representations of substrates and enzymes, thereby simulating their interaction patterns. It functions by processing the fundamental components of the attention mechanism, including the query (Q), key (K), and value (V) inputs, which facilitates a targeted analysis of specific input features. The modeling of the interaction process is as follows:

$$Q = f_Q(S); \quad K = f_K(E); \quad V = f_V(E)$$

$$\text{Interaction}_{S,E} = \text{CrossAttention}(Q, K, V)$$

$$= \text{softmax}\left(\frac{QK^T}{\sqrt{P/d}}\right)V$$

where Q is generated from the output of the molecule aggregator S, while K and V are derived from the output of the enzyme encoder E, using the projection functions $f = W^T x + b$ (where W represents the weight matrix and b is the bias vector). P denotes the embedding dimension, and d represents the number of attention heads.

The output section of the model correlates the comprehensive representations of substrates and enzymes with features induced by their interactions. Subsequently, these combined features are input into a multilayer, fully connected network to yield final predictions. The model generates an interaction label, indicating whether an interaction exists between a molecule and its corresponding enzyme target.

Implementation. Our model, developed using PyTorch,⁴⁷ underwent an extensive search for the optimal hyperparameters. The most effective combination was identified as a batch size of 512 and a learning rate of 5×10^{-5} . The training is conducted over 100 epochs, with an early stopping mechanism implemented to halt the training if no improvement is observed for 10 consecutive epochs. To mitigate overfitting, we used the Adam optimizer with a dropout rate of 0.2. In the compound feature extraction stage, the model is configured with three consecutive graph neural network layers. For binary classification, we employ the cross-entropy loss

function. The performance of our model is evaluated using several critical metrics, including the receiver operating characteristic area under the curve (AUROC), accuracy (ACC), Matthews correlation coefficient (MCC), and F1 score. The choice of these metrics ensures a balanced evaluation, preventing overfitting and providing a comprehensive view of the model's performance. The AUROC and ACC offer insights into the model's overall predictive capabilities, while the MCC and F1 score provide a nuanced understanding of its precision and recall, crucial for assessing the model's generalizability and robustness.

RESULTS AND DISCUSSION

Classifier Selection for Enzyme–Substrate Interaction Prediction. In the specialized domain of enzyme–substrate interaction prediction, the MEI model markedly surpasses other machine learning approaches in terms of crucial performance metrics. As shown in Table 1, the

Table 1. Performance of the MEI Models Based on Different Machine Learning Techniques (bold values indicate the highest performance metrics)

model	ACC	AUROC	MCC	F1 score
MEI _{-KNN}	0.931	0.961	0.802	0.867
MEI _{-RF}	0.940	0.977	0.835	0.867
MEI _{-XGB}	0.943	0.981	0.842	0.869
MEI _{-DNN}	0.965	0.987	0.897	0.919

MEI_{-DNN} (Deep Neural Network) variant particularly excels in accuracy, outperforming the MEI_{-RF} (Random Forest) and MEI_{-XGB} (XGBoost) models by about 2%. This increased accuracy is crucial, demonstrating the model's enhanced ability to accurately identify enzyme–substrate pairs. Furthermore, the MEI model excels in the AUROC metric, achieving a peak performance of 98.7%, which signifies its superior discriminatory power between interacting and noninteracting pairs. Additionally, the MEI model leads in both MCC and F1 scores, reflecting its balanced precision and recall capabilities. While these improvements may seem modest numerically, they are statistically significant, highlighting the MEI model's potential for practical applications. Beyond computational efficiency, the MEI model represents a transformative advancement in bioinformatics, paving the way for novel enzyme discoveries and optimizing biological pathway design. This innovation opens up new avenues in drug discovery and biocatalysis, marking a significant milestone in the field.

Comprehensive Performance Analysis of the MEI Model. We systematically compared the performance of the MEI model and a variety of benchmarking models, including the ESP-pre,²⁷ ESP,²⁷ ECFP-ESM,²⁷ and PMF-CPI²⁵ models. The ESP-pre and ESP models developed by Kroll et al.,²⁷ with the former performing a pretraining of the GNN to calculate task-specific numerical representations for the small molecules and the latter not, both applied a gradient-boosted decision tree model trained on the combined protein and small-molecule representations. The ECFP-ESM model utilizes Extended Connectivity Fingerprints (ECFP) for substrate representation and the ESM-1b model for enzyme representation, employing XGBoost to establish the classifier.²⁷ The PMF-CPI model,²⁵ originally trained on drug–target interaction data, was used to assess the reliability of using drug–target interaction models for the prediction of enzyme–

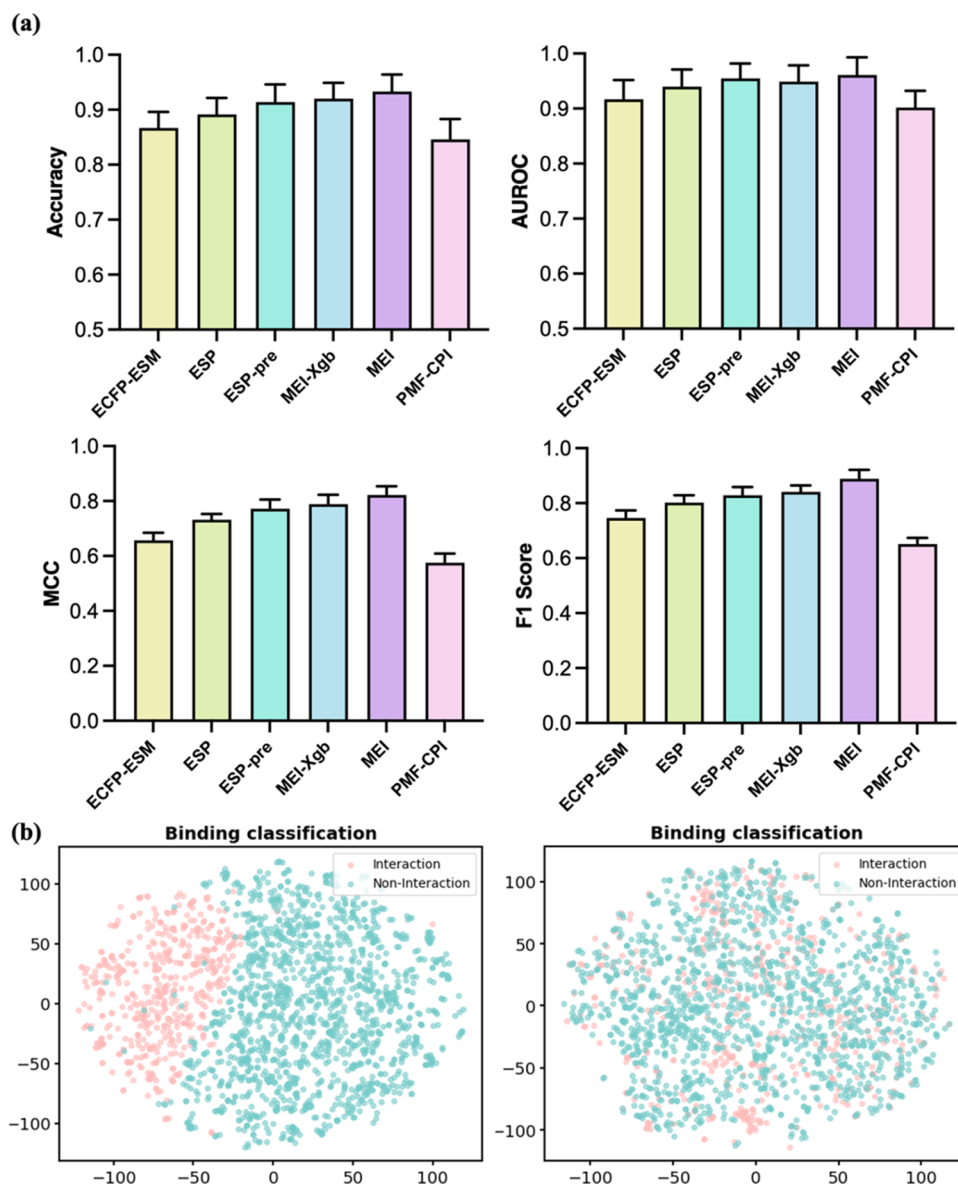


Figure 2. Performance comparison of the MEI and baseline models and t-SNE visualizations. (a) Comparative analysis of the MEI and baseline models in terms of four key metrics, including Accuracy, AUROC, MCC, and F1 Score. (b) t-SNE visualizations illustrating the feature spaces for the MEI model (left) and the ESP-pre model (right), highlighting the classes with Interaction (pink) and Non-Interaction (blue).

substrate interaction data, with testing conducted on the ESP model data set. The PMF-CPI model employs recursive neural networks to process protein embeddings based on the pretrained language model TAPE. It extracts molecular information from a graph encoder and generates output through a dense layer, exhibiting superior performance in binding affinity regression and CPI classification tasks compared to other notable methods.²⁵

In a thorough assessment of the ESP model data set, the performance of the MEI model stands out significantly against its competitors, highlighting its superiority in predictive modeling. This comparative analysis, focusing on four key performance metrics, reveals the MEI model's outstanding effectiveness and reliability, as shown in Figure 2a.

First, the MEI model achieves a remarkable accuracy of 93.4%, surpassing other models such as ECFP-ESM (86.7%) and ESP-pre (91.4%). This higher accuracy rate demonstrates the MEI model's superior precision in predictions. Regarding

the AUROC, an essential measure of a model's ability to differentiate between classes, the MEI model achieves an impressive score of 96.1%, outperforming its closest competitors, ESP and ESP-pre, and exceeding the baseline PMF-CPI model by a significant margin (71.4%).

Moreover, the MEI model's MCC score of 82.4% reflects its superior performance in binary classifications, standing well above other models, including ECFP-ESM (65.7%) and ESP-pre (77.2%). Finally, the MEI model leads the way with an F1 Score of 86.7%, reflecting its exceptional balance between precision and recall, which surpasses those of models such as ESP-pre (82.9%) and MEI-Xgb (84.1%). We included a comparison with the newly published ProSmith model⁴⁸ in the Supporting Information. Specifically, Table S1 presents the performance of our model when trained and tested on the same data set used by the ProSmith model. After fine-tuning, our model continued to achieve results comparable to those of the state of the art.

Notably, t-SNE visualizations in Figure 2b highlight the unique feature representations captured by the MEI model's neural network and the high-dimensional leaf index vectors from the ESP-pre's trained XGBoost classifier. On the left of the visualization, the MEI model's clusters exhibit a higher degree of separation compared to those of the ESP-pre model depicted on the right. This visual differentiation underlines the effectiveness of the MEI approach in capturing the nuances of enzyme–substrate interactions.

The PMF-CPI model, typically used for CPI predictions, has shown limitations when applied to direct enzyme–substrate interaction prediction. While it shares a similar architecture with the ESP model – leveraging the ESM2 protein language model to encode protein information and a GNN to encode substrate information, the expected outcomes were not achieved. As detailed in Supplementary Text 2, we further evaluated the performance of other CPI models,^{25,26} and the results indicated that retrained CPI models did not yield optimal predictions for enzyme–substrate interactions on both our custom enzyme–substrate test set and the standard ESP test set. Despite their strong performance in CPI prediction tasks, even after retraining on enzyme–substrate data sets, CPI models still failed to achieve satisfactory results in predicting enzyme–substrate interactions.

This finding highlights the fundamental differences between enzyme–substrate relationship prediction and traditional CPI models, which struggle to account for enzyme promiscuity and multifunctionality, making them less effective for accurately predicting enzyme–substrate interactions. This outcome underscores the specialized proficiency of the MEI model in this specific bioinformatics domain, further emphasizing its significant value.

Ablation Study. To further explore the significance of individual components, we devised these MEI variants:

MEI_{-Lite}: The interaction module is omitted, leaving MEI to rely solely on the drug and protein representations learned by the global and protein encoders for predicting enzyme–substrate interactions.

MEI_{-ESM_{1b}}: The CLEAN pretrained multilayer neural network is excluded, and MEI employs only standard evolutionary scale modeling (ESM-1b) for protein characterization.

MEI_{-GCN}: Graph Convolutional Networks (GCN), not restricted by chemical reactions, are used as molecular encoders.

Table 2 presents the experimental results of the drug library data set. The results from MEI_{-Lite} demonstrate the

Table 2. Results of the Ablation Experiments on the Test Set (bold values indicate the highest performance metrics)

model	ACC	AUROC	MCC	F1 score
MEI _{-GCN}	0.949	0.977	0.852	0.883
MEI _{ESM_{1b}}	0.952	0.981	0.860	0.889
MEI _{-Lite}	0.940	0.978	0.822	0.859
MEI	0.965	0.987	0.897	0.919

effectiveness of Attention in modeling interactions between enzyme proteins and small molecules. Furthermore, our findings indicate that using graph-convolution models for molecules without considering chemical reactions leads to a performance drop, suggesting that integrating catalytic reaction data into molecular characterization models enhances their

ability to learn enzyme representations related to substrate molecules. Moreover, the results for MEI_{-ESM_{1b}} reveal that a CLEAN-integrated, pretrained protein encoder to correlate protein sequence information with EC numbers can efficiently extract salient features. The superior performance of MEI underscores the advantage of using pretrained representations for both molecules and proteins, affirming the integrated approach is more effective in capturing complex biological interactions compared to the unoptimized representations in the baseline models.

Analysis of Experimental Results of Different Test Sets. To rigorously evaluate the robustness, generalizability, and applicability of the MEI model in real-world scenarios, we carefully curated two different test sets. The first comprises over 1000 enzyme–substrate reaction pairs, featuring meticulously curated experimental data on oxidoreductases, hydrolases, and lyases, along with their respective substrates. It is important to highlight that the protein sequences in this data set share less than 60% similarity with those in the training set, and all substrate molecules were excluded during the model's initial training and validation phases. This approach ensures a fair and unbiased assessment of the model's performance. The second data set, equally diverse, comprises over 1000 experimental enzyme–substrate pairs from the Esterase²¹ and DUF849⁴⁹ enzyme families, compiled from various literature sources. This data set was also excluded from the MEI model's initial training. Derived from high-throughput enzyme screening and backed by comprehensive structural, modeling, genomic, and metabolic analyses, these independent external data sets provide a robust platform for validating the MEI model's performance, demonstrating its applicability not only in controlled environments but also in complex real-world situations.

The performance of the MEI model on these data sets is significant. As seen in Figure 3a,b, it surpasses the RF and XGB models constructed using the same molecular and enzyme characterization methods as MEI. Furthermore, it outperforms the RF_{-esp}, ESP, MEI_{-esp} models trained on the data sets used by the ESP models, as well as the GNN_{-mb} model constructed using the molecular GNN method introduced by Kroll et al.²⁷ The GNN_{-mb}, MEI, RF, and XGB models were trained on the MEI data set, while the RF_{-esp}, ESP, and MEI_{-esp} models were trained on the ESP model data set. RF_{-esp} relied on the traditional molecular fingerprint ECFP for characterization, while the protein characterization was handled by the random forest classifier output from the ESM model. This demonstrates the generalization ability of traditional machine learning methods. In Figure 3a, the MEI model, representing the first test set, outperforms all competitors across all metrics, particularly achieving the highest accuracy and MCC. Notably, the MEI model's AUROC value exceeds that of the current state-of-the-art GNN_{-mb} and ESP models by at least 28.9%, indicating its superior prediction accuracy. In Figure 3b, MEI's advantage is evident for the second test set, particularly in terms of AUROC and MCC, where it outperforms the RF model by 16.3 and 5.3%, respectively. These results emphasize the robustness and enhanced predictive power of the MEI model compared to those of the underperforming RF and ESP models.

The comparison with GNN_{-mb} further illustrates that the model trained on the MEI data set generally outperforms that trained on the ESP data set, affirming the reliability and generalization of our data set. When comparing the models

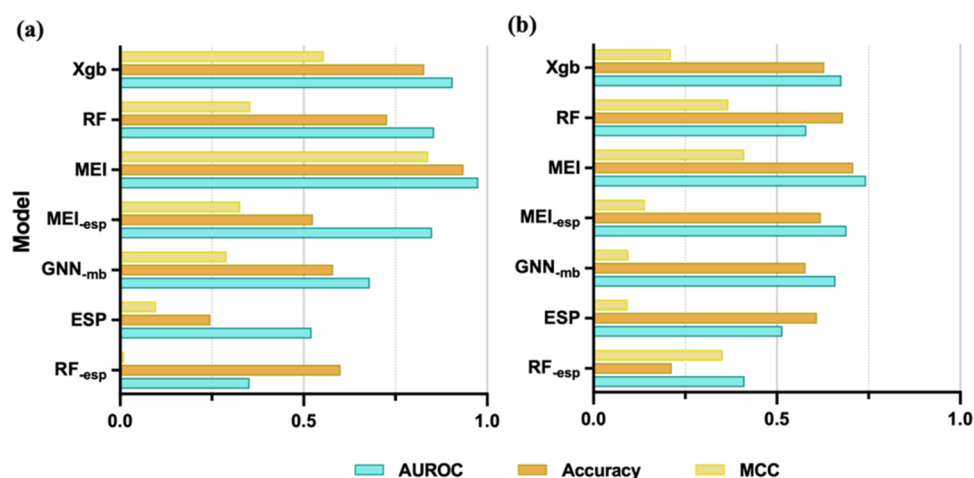


Figure 3. Model performance comparison on the (a) first and (b) second enzyme test sets in terms of AUROC, accuracy, and MCC. The colored bars represent different evaluation metrics.

trained on the ESP data sets, the consistently superior test results of the MEI model highlight its rational overall design and superior generalization capabilities. This visualization succinctly confirms the strengths of the MEI model, communicating its superior accuracy and overall prediction quality clearly and concisely.

The comprehensive analysis across various test sets firmly establishes the MEI model as a leading tool in enzymatic modeling and analysis. Its ability to perform consistently and accurately across different enzyme families and complex scenarios sets a new benchmark for future developments in this field. The stark contrast between the ESP model's limitations and the MEI model's consistent outperformance of its variants further underscores the necessity for advanced models like MEI in biological studies. Applying this strategy to other enzyme families holds immense potential to significantly enhance our understanding of the enzymatic landscape.

Fine-Tuning on CYP Inhibitor and Substrate Data.

The results presented above clearly demonstrate the remarkable performance of our model in classifying enzyme–substrate interaction benchmarks. In this section, we aim to further refine the model's predictive capabilities by fine-tuning it on CYP inhibitor and substrate data sets for drug selectivity prediction.

The human CYP family has 57 isoforms,⁵⁰ and CYPs are the primary enzymes responsible for drug metabolism, accounting for approximately 75% of overall metabolism, with about 95% of this activity being attributed to five isoforms: 1A2, 2C9, 2C19, 2D6, and 3A4.⁵¹ The CYP-mediated metabolism of novel chemical entities is crucial in drug development because it can significantly influence the compound's initial bioavailability, bioactivity, and safety profile.⁵² CYP inhibitors can affect the metabolism of drugs by reducing the activity of CYP enzymes involved, thereby modifying the pharmacokinetics and pharmacodynamics of the drugs. On the other hand, the rate and extent of CYP substrate metabolism can be influenced by the presence of CYP inhibitors or other factors. Predicting the potential of a drug as a CYP inhibitor or substrate is a complex process that involves many factors and variables, such as an individual's genetic makeup, age, and overall health. As such, predictions may not always be accurate and may need to be confirmed through further testing and analysis. There have been a lot of studies focusing on the prediction of inhibitors or

substrates for 5 major CYPs (1A2, 2C9, 2C19, 2D6, and 3A4), such as DeepCYP,⁵³ SuperCYPsPred,⁵⁴ CYplebrity,⁵⁵ iCYP-MFE,⁵⁶ and others.^{57–59}

Case study of CYP Inhibitors. Compared to developing a model from scratch, fine-tuning existing models for selective prediction not only saves time but also achieves commendable predictive accuracy with a limited data set. In this study, the fine-tuning process utilized the training set from the iCYP-MFE model,⁵⁶ and evaluation was conducted on the same test set used by the baseline model. Importantly, neither the training nor the testing data used for fine-tuning were present in the original MEI model's training data. The initial chemical data were obtained from the PubChem BioAssay Database.⁶⁰ The samples were categorized as “active” (inhibitors) or “inactive” (noninhibitors) against CYP enzymes. The curated sample sizes varied across different isoforms, specifically, 11,357 for 1A2, 11,244 for 2C9, 11,871 for 2C19, 12,284 for 2D6, and 13,711 for 3A4. Notably, the training and validation data sets remained independent.

To fairly assess the performance of our fine-tuned MEI model, we benchmarked it against Swiss-ADME,⁶¹ a highly reliable prediction framework developed by Daina et al. in 2017, SuperCYPsPred,⁶² the most recent prediction framework developed by Banerjee et al. in early 2020, and iCYP-MFE,⁵⁶ a model published in 2022 for predicting human CYP inhibitors using a multitask learning approach.

As shown in Table 3, our MEI model significantly outperforms the Swiss-ADME, iCYP-MFE, and SuperCYPsPred models in most categories. Additionally, we compared our model with ADMET prediction tools such as admetSAR 2.0⁶³ and the newly released ADMETlab 3.0.⁶⁴ The results demonstrate that, for the CYP1A2 and CYP2C9 isoforms, our MEI model's prediction ACC and AUROC are very close to the best-performing Swiss-ADME model and ADMETlab 3.0 prediction tools. Furthermore, for several other isoforms, the MEI model exhibits the highest ACC and superior MCC scores, indicating a more balanced and reliable predictive performance, especially in scenarios with unbalanced data. This also highlights its efficiency in accurately predicting enzyme interactions in complex isoforms. Table S4 also compares the test results of the MEI model without fine-tuning. These results illustrate the robustness of the MEI model in delivering high-precision predictions across a wide

Table 3. Performance of the Finely Tuned MEI, iCYP-MFE, Swiss-ADME, and SuperCYPsPred Models (bold values indicate the highest performance metrics)

isoform	model	ACC	AUROC	MCC
CYP1A2	Swiss-ADME	0.88		0.75
	SuperCYPsPred	0.69	0.83	0.46
	iCYP-MFE	0.82	0.91	0.64
	admetSAR 2.0	0.81	0.88	
	ADMETlab3.0	0.86	0.94	0.73
	MEI	0.86	0.92	0.72
CYP2C9	Swiss-ADME	0.75		0.47
	SuperCYPsPred	0.61	0.87	0.38
	iCYP-MFE	0.78	0.89	0.58
	admetSAR 2.0	0.80	0.86	
	ADMETlab3.0	0.85	0.92	0.68
	MEI	0.84	0.91	0.67
CYP2C19	Swiss-ADME	0.82		0.62
	SuperCYPsPred	0.64	0.90	0.40
	iCYP-MFE	0.78	0.86	0.57
	admetSAR 2.0	0.81	0.87	
	ADMETlab3.0	0.78	0.83	0.55
	MEI	0.84	0.91	0.67
CYP2D6	Swiss-ADME	0.71		0.32
	SuperCYPsPred	0.81	0.88	0.53
	iCYP-MFE	0.80	0.93	0.64
	admetSAR 2.0	0.86	0.84	
	ADMETlab3.0	0.87	0.88	0.58
	MEI	0.88	0.87	0.54
CYP3A4	Swiss-ADME	0.79		0.57
	SuperCYPsPred	0.77	0.88	0.56
	iCYP-MFE	0.80	0.89	0.60
	admetSAR 2.0	0.65	0.85	
	ADMETlab3.0	0.83	0.91	0.67
	MEI	0.84	0.91	0.65
overall	MEI	0.85	0.91	0.67

range of subtypes, substantially surpassing existing methodologies. The fine-tuned MEI model's ability to directly predict interactions between all CYP isoforms and small molecules, providing a comprehensive overview, marks a significant advancement over other multitask-based models. This capability is critical in predicting drug metabolism and interactions, offering invaluable insights into pharmacological research and development.

Case Study of CYP Substrates. Here, we conducted a comparative analysis of our fine-tuned MEI model against the CYP substrate prediction functions of several comprehensive ADMET (absorption, distribution, metabolism, excretion, and toxicity) prediction tools, including Interpretable-ADMET,⁶⁵ ADMETlab 2.0,⁶⁶ admetSAR 2.0,⁶³ and the newly released ADMETlab 3.0.⁶⁴ The evaluation is based on the data set, originally curated by Zhang et al.⁶⁷ This data set contains 9,233 substrates and the corresponding CYP isomers. Moreover, the data set presents a blend of both labeled and unlabeled data, offering a comprehensive resource for subsequent analysis. All compounds in the data set were converted into canonical SMILES strings using RDKit.

The analysis, as illustrated in Table 4, underscores the fine-tuned MEI model as a superior predictive tool for CYP substrates across various isoforms. It outperforms other models in terms of accuracy, AUROC, and MCC metrics. Notably, the MEI model achieves accuracy approximately 10% higher than

Table 4. Performance of Fine-Tuned MEI and Other Publicly Accessible ADMET Tools (bold values indicate the highest performance metrics)

isoform	model	ACC	AUROC	MCC
CYP1A2	ADMETlab2.0	0.82	0.84	0.63
	ADMETlab3.0	0.78	0.85	0.56
	Interpretable-ADMET			
	admetSAR 2.0			
CYP2C9	MEI	0.90	0.92	0.71
	ADMETlab2.0	0.74	0.79	0.41
	ADMETlab3.0	0.73	0.78	0.45
	Interpretable-ADMET	0.76	0.68	0.40
CYP2C19	admetSAR 2.0	0.78	0.63	
	MEI	0.90	0.90	0.60
	ADMETlab2.0	0.74	0.93	0.50
	ADMETlab3.0	0.80	0.88	0.58
CYP2D6	Interpretable-ADMET			
	admetSAR 2.0			
	MEI	0.90	0.93	0.61
	ADMETlab2.0	0.66	0.78	0.33
CYP3A4	ADMETlab3.0	0.75	0.84	0.51
	Interpretable-ADMET	0.80	0.75	0.55
	admetSAR 2.0	0.78	0.77	
	MEI	0.89	0.94	0.73
overall	ADMETlab2.0	0.75	0.80	0.51
	ADMETlab3.0	0.76	0.84	0.53
	Interpretable-ADMET	0.70	0.68	0.42
	admetSAR 2.0	0.66	0.70	
overall	MEI	0.90	0.95	0.79
	MEI	0.90	0.92	0.71

that of its closest competitors, ADMETlab 2.0/3.0, along with significant enhancements in AUROC and MCC scores. This consistency across various isoforms highlights the robustness and superiority of MEI as the leading predictive model in this study.

Dual-Target Affinity Prediction for CYP3A4/CYP3A7 Substrate Molecules. Xu et al. assayed almost 5000 drugs and drug-like compounds toward CYP3A4 and CYP3A7 via quantitative high-throughput screening and identified a number of multitarget or selective inhibitors. Based on these data, they constructed machine learning models to detect CYP3A4 or CYP3A7 selective inhibitors.⁶⁸ However, without incorporating target-specific information, they have to build separate models for each target and also did not develop regression models with activity data.

We further fine-tuned our model on this CYP3A4/CYP3A7 selectivity data set for regression tasks. As depicted in Figure 4a, our model proficiently predicts affinity and inhibitory relationships, even for just 30% of the samples. A comparative analysis with PMF-CPI validates that the fine-tuned MEI rivals current leading methodologies. Our model proficiently predicts affinity and inhibitory relationships, even with just 30% of the samples. Despite sharing 94.4% sequence similarity,²⁵ CYP3A4 and CYP3A7, both belonging to the human CYP3A subfamily, exhibit varied inhibitory affinities toward certain compounds, as shown in Figure 4b. The pIC₅₀ values of these molecules show a significant activity difference of almost 1 order of magnitude between CYP3A4 and CYP3A7. Our model accurately predicts the affinities of such compounds with an error margin around 0.1. Accurate predictions of affinity values and binary interaction relationships are not mutually exclusive

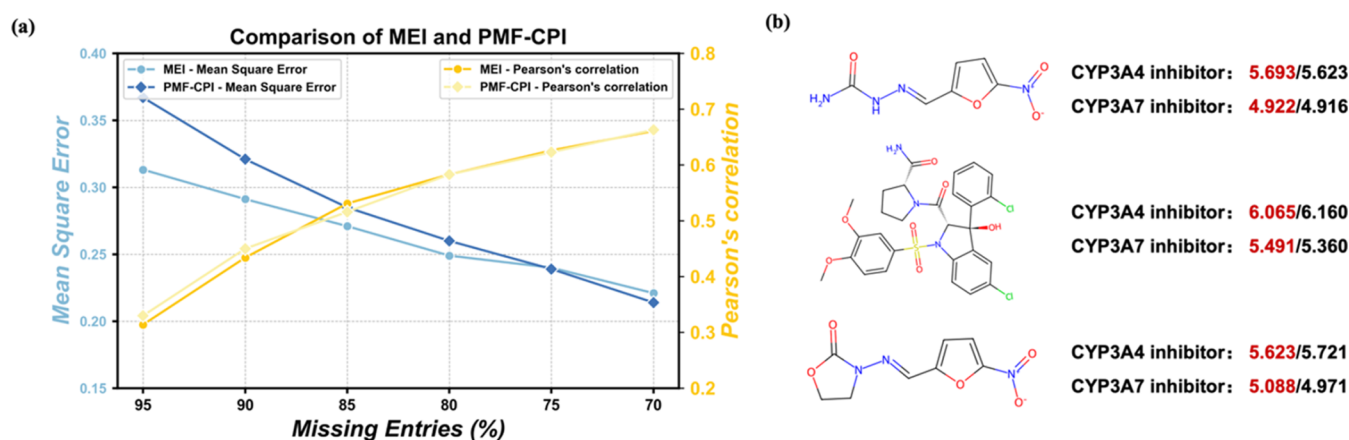


Figure 4. MEI model fine-tuning on CYP3A4/CYP3A7 selectivity data sets. (a). Plots of Mean Squared Error (MSE) and Pearson correlation across regression sets with 70–95% data availability were performed, along with a comparative analysis with PMF-CPI. (b). Illustrates inhibitors, contrasting predicted (red) with actual pIC₅₀ values (black) for CYP3A4/CYP3A7.

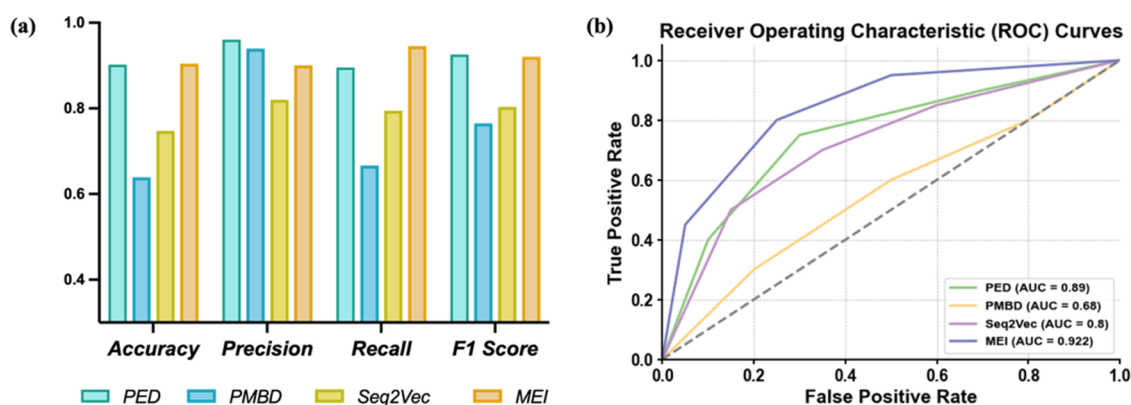


Figure 5. Performance comparison of MEI with sequence-based protein classification models in terms of (a) evaluation metrics including accuracy, precision, recall, and F1 score, and (b) receiver operating characteristic (ROC) curve with AUC values signifying the discriminative capacity of each model.

but rather complementary. When enzyme–substrate pairs are introduced into our computational framework, the integrated approach utilizing both regression and classification models facilitates a more nuanced identification of selection relationships. This dual-model methodology offers robust insights into complex biochemical interactions, enhancing our ability to predict how variations in molecular structures impact binding affinity and inhibitory potency.

Enzymatic Plastic Degradation Analysis within Green Chemistry. The escalating environmental impact of global plastic usage, compounded by the challenges of improper waste disposal, has underscored the need for innovative recycling solutions.^{69,70} The exploration for enzymes that can efficiently degrade plastics has intensified, driven by the promise of sustainable waste management. However, the vast array of enzyme sequences presents a significant challenge in identifying candidates with the requisite degradation capabilities. To address this, we optimized our MEI model to assess its predictive performance and applicability in the field of green chemistry, specifically targeting the degradation of various plastic types.

The PED model⁷¹ employs machine learning to interpret enzyme amino acid sequences for predicting plastic degradation, transforming them into computational representations via sequence learning and enriching them with features from both amino acid and global sequence data. This comprehensive

framework, which combines validated enzymes and plastic substrates into a contextual data set, outperforms conventional sequence-based protein classifiers with a 90.2% accuracy, marking a state-of-the-art achievement. In contrast, PMBD⁷² utilizes convolutional neural networks to analyze one-hot encoded sequences for degradation predictions, and Seq2Vec⁷³ adopts an unsupervised learning method to derive latent sequence representations. Both models' outputs are fed into an XGBoost classifier, maintaining methodological consistency for fair performance comparison. This synergy of varied approaches underscores PED's advancement in the nuanced field of enzyme-related plastic degradation.

By employing a transfer learning approach, we fine-tuned our model, pretrained on the MEI data set, on the data set constructed by PED. Unlike the aforementioned models, we used the SMILES representations of monomers from different plastics to denote these types. The subsequent molecular and enzyme sequence inputs were processed in the same manner as for prior training. Ultimately, we compared our prediction results with those of PED, Seq2Vec, and PMBD across various metrics (Figure 5). Figure 5a shows that the MEI model's performance across various metrics (accuracy, precision, recall, F1 score) is on par with the domain-specific SOTA model, PED. Table S5 also compares the test results of the MEI model without fine-tuning. This indicates that our model has successfully transferred knowledge from the MEI data set

and has adapted well to the new domain represented by the PED data set. Figure 5b reveals a 3% increase in the MEI model's AUROC (Area Under the Receiver Operating Characteristic curve) compared to PED. This improvement in the AUROC score is significant because it suggests better overall performance in classifying positive and negative classes across various thresholds.

In summary, the transfer learning approach has not only preserved high performance but also enhanced the model's ability to distinguish between classes, as reflected by the improved AUROC. This demonstrates that the MEI universal model, pretrained on a full type of enzyme–substrate data set, is highly effective and can be fine-tuned to adapt to various enzyme-related tasks, serving as a reliable tool for its designated purposes while maintaining cutting-edge performance and providing better generalization.

CONCLUSIONS

The MEI model unveiled in this study emerges as a pioneering machine learning methodology for predicting enzyme–substrate interactions. It integrates data from enzymatic reactions, going beyond traditional molecular characterization, thereby enhancing its analytical power and providing comprehensive insights into the enzyme–substrate interplay. With exceptional accuracy and multifunctionality, the MEI model outperforms existing models in various testing environments. Fine-tuning of the CYP inhibitor and substrate data further refines its predictive capabilities, demonstrating superior performance in classifying enzyme–substrate interaction benchmarks. Moreover, through transfer learning in enzymatic plastic degradation analysis within green chemistry, the MEI model achieves performance comparable to the domain-specific state-of-the-art model, PED, with a 3% increase in AUROC, indicating improved performance in classifying positive and negative classes. However, challenges remain in enhancing the model's interpretability, particularly regarding its reliance on pretrained protein representations, and in fine-tuning for specific enzyme families due to data heterogeneity and scarcity.

The model's success underscores the potential of machine learning to dissect complex biochemical interactions, which is not only set to revolutionize drug discovery but also to extend its applications into areas like green chemistry and biocatalysis. The prospect of expanding and refining the MEI model with future database updates promises to sharpen its utility and bolster our understanding of metabolic pathways and enzyme mechanics. Such continuous improvements will likely yield a surge in efficiency and open new avenues in scientific research, suggesting a breadth of impacts that transcends biopharmaceuticals. In essence, the MEI model, while currently limited by the extent of available data, hints at a future of prolific growth, where it will stand as an indispensable tool in advancing the field of bioinformatics and beyond.

ASSOCIATED CONTENT

Data Availability Statement

The trained models, code, processed data, and enzyme–substrate interaction data set supporting the results reported in this article are publicly available at <https://github.com/KeeliaQWJ/MEI> and the Zenodo repository at 10.5281/zenodo.13884790. The data set includes the enzyme–substrate pairs and the trained MEI model used for predictions. The original data were retrieved from the Rhea database, which can

be accessed at <https://www.rhea-db.org>. The method for generating negative samples can be found at <https://github.com/rxn4chemistry/biocatalysis-model>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.4c01801>.

Additional comparative analyses, detailed performance metrics, and evaluations of model fine-tuning; tables providing quantitative performance metrics and comparative analyses for the model across various tasks, highlighting its effectiveness relative to baseline approaches (PDF)

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Notes

The authors declare no competing financial interest.

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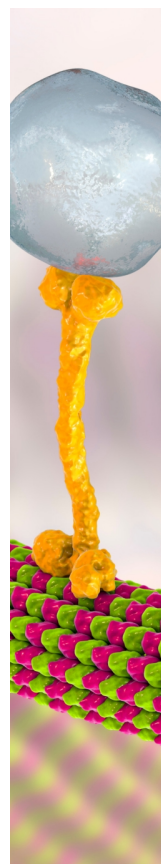
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